BAYESIAN MODELING OF COMPLEXITY

Bayesian Linear Models

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Agenda

1 Recap & Today

2 Bayesian Linear Models

- Linear Regression
- Maximum Likelihood vs. Ridge Regression

3 Model Quality

- PPC & Diagnostics
- Model Comparison

4 Wrap-up & Next

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Last Week Recap

What we covered in Session 6:

- MCMC: Sampling from intractable posteriors
- Metropolis-Hastings algorithm
- MCMC diagnostics: $\hat{R}$, ESS, trace plots, divergences
- PyMC workflow
- Sampling-based Bayesian calibration

Today's focus:

Bayesian regression

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Why Bayesian Linear Regression?

Linear regression is everywhere:

- Predicting outcomes from covariates
- Understanding relationships between variables

Bayesian approach advantages:

- Full uncertainty quantification (not just point estimates)
- Incorporates prior knowledge (regularization)
- Natural framework for hierarchical structure
- Predictions with credible intervals

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Linear Regression: Setup

Goal: Model the relationship between data and parameters as linear:

$$\mathbf{y} = \mathbf{X}\beta + \epsilon,$$

where

$\mathbf{y}_{[n\times 1]}$	dependent variable (outcome)
$\mathbf{X}_{[n\times k]}$	design matrix (covariates)
$\beta_{[k\times 1]}$	coefficients (parameters)
$\epsilon_{[n\times 1]}$	noise vector (residuals)
$n \times k$	# of data points × # of covariates

Standard assumption: $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ (i.i.d. Normal noise)

This implies: $y_i \mid \mathbf{x}_i, \beta, \sigma \sim \mathcal{N}(\mathbf{x}_i^T \beta, \sigma^2)$

Two approaches: Frequentist (OLS, MLE) vs. Bayesian (prior + posterior)

Example: Simple Linear Regression

Model: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$

Matrix form:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

Interpretation:

- β_0: Intercept (expected y when $x = 0$)
- β_1: Slope (change in y per unit change in x)
- σ^2: Residual variance (unexplained variation)

Frequentist solution (OLS): $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$

Bayesian solution: Posterior distribution over $\boldsymbol{\beta}$ and σ

Bayesian Regression: Three Ingredients

To do Bayesian regression, we need to specify:

1. **Prior:** $P(\beta, \sigma)$
   - What do we believe before seeing data?
   - Can encode domain knowledge or regularization
2. **Likelihood:** $P(\mathbf{y} \mid \mathbf{X}, \beta, \sigma)$
   - Probability of observed data given parameters
   - Usually: $\mathcal{N}(\mathbf{X}\beta, \sigma^2 I)$
3. **Posterior:** $P(\beta, \sigma \mid \mathbf{y}, \mathbf{X})$
   - Updated beliefs after seeing data
   - From Bayes rule: $P(\beta, \sigma \mid \mathbf{y}, \mathbf{X}) \propto P(\mathbf{y} \mid \mathbf{X}, \beta, \sigma) P(\beta, \sigma)$
   - Compute via MCMC (Session 6!)

Today's focus: Understanding these three components and their connections

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Case 1: Uniform Prior (Maximum Likelihood)

Prior: $P(\beta) = \frac{1}{b-a}$ (uniform distribution: $\beta \sim \mathcal{U}(a, b)$)

Likelihood: Assuming $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2)$ are independent:

$$P(\mathbf{y} \mid \mathbf{X}, \beta) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma_\epsilon^2}} \exp\left[-\frac{1}{2\sigma_\epsilon^2}(y_i - \mathbf{x}_i^T\beta)^2\right]$$

Posterior: $P(\beta \mid \mathbf{y}, \mathbf{X}) \propto P(\beta)P(\mathbf{y} \mid \mathbf{X}, \beta)$

$$\propto \frac{1}{b-a}\prod_{i=1}^{n} \exp\left[-\frac{1}{2\sigma_\epsilon^2}(y_i - \mathbf{x}_i^T\beta)^2\right] \quad \propto \exp\left[-\frac{1}{2\sigma_\epsilon^2}\sum_{i=1}^{n}(y_i - \mathbf{x}_i^T\beta)^2\right]$$

Result: Maximizing posterior $\Leftrightarrow$ Minimizing $\sum(y_i - \mathbf{x}_i^T\beta)^2$

$\boxed{\text{Uniform prior} \Rightarrow \text{Maximum Likelihood Estimation (MLE)}}$

Case 2: Gaussian Prior (Ridge Regression)

Prior: $P(\beta) \propto \exp\left[-\frac{1}{2\sigma_0^2}\|\beta - \mu_0\|^2\right] \quad (\beta \sim \mathcal{N}(\mu_0, \sigma_0^2 I))$

Likelihood: $P(\mathbf{y} \mid \mathbf{X}, \beta) \propto \prod_{i=1}^n \exp\left[-\frac{1}{2\sigma_\epsilon^2}(y_i - \mathbf{x}_i^T\beta)^2\right]$

Posterior: $P(\beta \mid \mathbf{y}, \mathbf{X}) \propto P(\beta)P(\mathbf{y} \mid \mathbf{X}, \beta)$

$$\propto \exp\left[-\frac{1}{2\sigma_0^2}\|\beta - \mu_0\|^2\right] \prod_{i=1}^n \exp\left[-\frac{1}{2\sigma_\epsilon^2}(y_i - \mathbf{x}_i^T\beta)^2\right]$$

$$\propto \exp\left[-\frac{1}{2}\left(\frac{1}{\sigma_0^2}\|\beta - \mu_0\|^2 + \frac{1}{\sigma_\epsilon^2}\sum_{i=1}^n (y_i - \mathbf{x}_i^T\beta)^2\right)\right]$$

MAP (Maximum a-posteriori):

$$\hat{\beta}_{\text{MAP}} = \arg\min_\beta \left[\sum_{i=1}^n (y_i - \mathbf{x}_i^T\beta)^2 + \frac{\sigma_\epsilon^2}{\sigma_0^2}\|\beta - \mu_0\|^2\right]$$

$$\boxed{\text{Gaussian prior} \Leftrightarrow \text{Ridge Regression (L2 penalty)}}$$

Regularization as Priors

Key insight: Priors = Regularization

Gaussian prior: $\beta \sim \mathcal{N}(\mathbf{0}, \tau^2 I)$

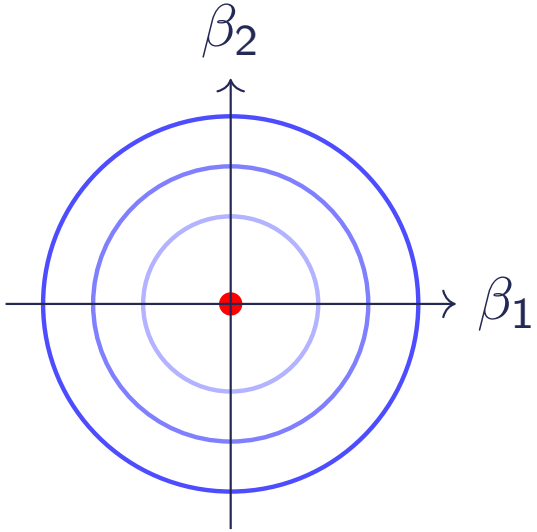
$\Leftrightarrow$ Ridge penalty $\frac{1}{2\tau^2}\|\beta\|_2^2$ in MAP

- Prior scale τ controls shrinkage strength
- $\downarrow \tau$: stronger shrinkage $\rightarrow$ simpler model
- $\uparrow \tau$: weaker shrinkage $\rightarrow$ closer to OLS
- Choose τ via: cross-validation, domain knowledge, fully Bayesian (hyperprior on τ)

Other priors:

- Student-t prior $\rightarrow$ Robust to outliers

L2 penalty contours:



Ridge regularization

Contours show equal prior probability, shrinks toward origin

Full Bayesian Treatment

Beyond MAP: Don't just find maximum, get full posterior!

Model:

$$\begin{aligned} \beta &\sim \mathcal{N}(\mu_0, \sigma_0^2 I) && \text{(prior on coefficients)} \\ \sigma &\sim \text{HalfCauchy}(5) && \text{(prior on noise)} \\ y_i \mid \mathbf{x}_i, \beta, \sigma &\sim \mathcal{N}(\mathbf{x}_i^T \beta, \sigma^2) && \text{(likelihood)} \end{aligned}$$

Posterior: $P(\beta, \sigma \mid \mathbf{y}, \mathbf{X})$ (via MCMC)

What we get:

- Full distribution, not just point estimate
- **Credible intervals** for each β_j: "95% probability $\beta_j \in [a, b]$"
- Predictions with uncertainty: $P(y_{\text{new}} \mid x_{\text{new}}, \mathbf{y}, \mathbf{X})$
- Can answer questions like: "What's probability that $\beta_1 > 0$?"

Analytical vs. Computational

Conjugate case (rare):
If $\beta \sim \mathcal{N}(\mu_0, \Sigma_0)$ and likelihood is Normal, posterior is also Normal:

$$\beta \mid \mathbf{y} \sim \mathcal{N}(\mu_n, \Sigma_n)$$

where

$$\Sigma_n^{-1} = \Sigma_0^{-1} + \frac{1}{\sigma^2}\mathbf{X}^T\mathbf{X}$$

$$\mu_n = \Sigma_n(\Sigma_0^{-1}\mu_0 + \frac{1}{\sigma^2}\mathbf{X}^T\mathbf{y})$$

But: Only if σ is known!

General case (MCMC):
For realistic models:

- Unknown σ (non-conjugate)
- Non-Normal likelihood (e.g., logistic)
- Hierarchical structure
- Complex priors

MCMC Challenge:

- Ensuring correct sampling via **diagnostics**!

Practical approach: We always use PyMC unless we need analytical results

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Posterior Predictive Checks (PPC)

Question: Does our model generate realistic data?

Posterior predictive distribution:

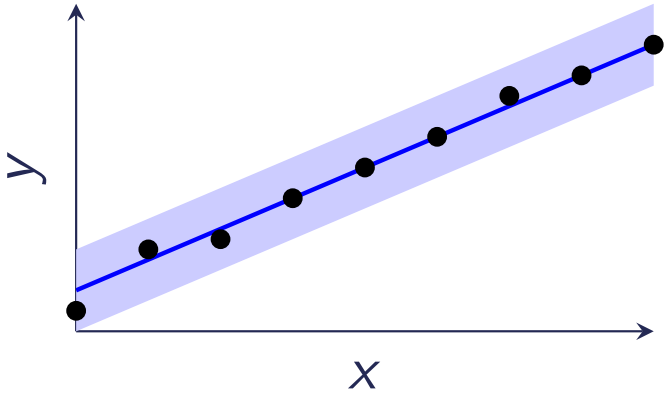
$$p(y^* \mid \mathbf{X}, \mathbf{y}) = \int p(y^* \mid \mathbf{X}, \beta, \sigma) p(\beta, \sigma \mid \mathbf{y}, \mathbf{X}) d\beta d\sigma$$

Process:

1. Draw $(\beta^{(s)}, \sigma^{(s)})$ from posterior
2. Generate $y^{*(s)} \sim \mathcal{N}(\mathbf{X}\beta^{(s)}, \sigma^{(s)})$
3. Repeat for many samples
4. Compare to observed $\mathbf{y}$

PyMC makes this very easy!

Prediction bands:



Blue band: 95% posterior predictive interval

Visual checks:

- Do replicated data look like observed?
- Check residuals: mean 0, constant variance
- Coverage: Do 95% bands contain 95% of data?

Test Statistics for PPC

Idea: Compare summary statistics of observed vs. replicated data

Common test statistics:

- $T(\mathbf{y}) = \text{mean}(\mathbf{y})$: central tendency
- $T(\mathbf{y}) = \text{var}(\mathbf{y})$: spread
- $T(\mathbf{y}) = \max(\mathbf{y})$: extreme values
- $T(\mathbf{y}) = \text{skewness}(\mathbf{y})$: asymmetry
- $T(\mathbf{y}) = \chi^2$: goodness-of-fit

Bayesian p-value: $p_B = P(T(\mathbf{y}^{\text{rep}}) \geq T(\mathbf{y}) \mid \mathbf{y})$

Proportion of replicated datasets with more extreme statistic

Interpretation:

- $p_B \approx 0.5$: observed statistic typical of model
- p_B very small (< 0.05) or large (> 0.95): potential misfit

Caution: Not a classical p-value! Just a diagnostic tool.

P-Value: Example

Two models are fitted to a longitudinal dataset of about 2000 Australian adolescents whose smoking patterns were recorded every six months (via questionnaire) for a period of three years. Interest lay in the extent to which smoking behavior could be predicted based on parental smoking and other background variables, and the extent to which boys and girls picked up the habit of smoking during their teenage years.

Test variable	$T(y)$	Model 1		Model 2	
		95% int. for $T(y^{\text{rep}})$	p-value	95% int. for $T(y^{\text{rep}})$	p-value
% never-smokers	77.3	[75.5, 78.2]	0.27	[74.8, 79.9]	0.53
% always-smokers	5.1	[5.0, 6.5]	0.95	[3.8, 6.3]	0.44
% incident smokers	8.4	[5.3, 7.9]	0.005	[4.9, 7.8]	0.004

Table 1: Posterior predictive checks for two models.

Residuals

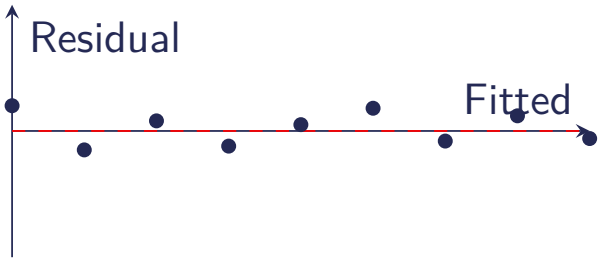
Residual analysis:

- Plot residuals $e_i = y_i - \mathbb{E}[y_i \mid \text{data}]$ (difference btw. observed & predicted)
- Check: mean 0, constant variance (residual should be random noise if the model is correct)
- Look for patterns (non-linearity, heteroscedasticity)

Influential points:

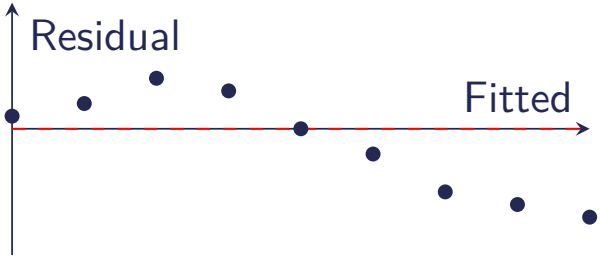
1. How unusual is $\mathbf{x}_i$?
2. How unusual is y_i given $\mathbf{x}_i$?
3. How much does removing i change fit?

Good residual plot:



Random scatter around zero

Bad residual plot:



Pattern $\rightarrow$ model misspecification

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The Model Selection Problem

Scenario: Multiple candidate models for the same data

Examples:

- Linear vs. quadratic vs. cubic
- Different sets of predictors
- Different prior specifications
- Different likelihood families (Normal vs. Student-t)

Question: Which model is "best"?

Two competing goals:

1. **Fit:** Model should explain observed data well
2. **Simplicity:** Avoid overfitting

Solution: Information criteria that balance fit and complexity

- WAIC (Widely Applicable Information Criterion)
- LOO-CV (Leave-One-Out Cross-Validation)

WAIC: Widely Applicable Information Criterion

Definition: $\text{WAIC} = \text{lppd} - p_{\text{WAIC}}$

$$\text{lppd} = \sum_{i=1}^{n} \log \int p(y_i \mid \boldsymbol{\theta}) p(\boldsymbol{\theta} \mid \mathbf{y}) d\boldsymbol{\theta} \quad \text{(log pointwise predictive density)}$$

$$p_{\text{WAIC}} = \sum_{i=1}^{n} \text{Var}_{\text{post}}[\log p(y_i \mid \boldsymbol{\theta})] \quad \text{(effective number of parameters)}$$

- **lppd**: How well the model predicts *each* data point (averaged over posterior)
- p_{WAIC}: Complexity penalty (higher flexibility $\rightarrow$ larger penalty)
- **WAIC**: Balance between fit and complexity

Higher WAIC $\Rightarrow$ better predictive performance

Properties:

- Fully Bayesian (uses entire posterior)
- Works for any model & easy to compute in PyMC

LOO-CV: Leave-One-Out Cross-Validation

Idea: How well does the model predict unseen data?

Leave-one-out cross-validation:

1. Leave out observation i
2. Fit model to remaining data $\mathbf{y}_{-i}$
3. Predict left-out y_i
4. Repeat for all $i = 1, \ldots, n$
5. Sum log predictive densities

Definition

$$\text{LOO} = \sum_{i=1}^{n} \log p(y_i \mid \mathbf{y}_{-i})$$

Problem: Need to refit model n times $\rightarrow$ expensive!

Higher LOO $\Rightarrow$ **better** (Similar to WAIC)

Note: Some softwares report $-2 \times LOO$ or $-2 \times WAIC$, which is due to standardization of information metrics (?).

Interpreting WAIC and LOO

Both provide:

- Point estimate of predictive performance
- Standard error (uncertainty in estimate)

How to compare models:

1. Compute WAIC (or LOO) for each model
2. **Difference in WAIC/LOO:** $\Delta\text{WAIC} = \text{WAIC}_1 - \text{WAIC}_2$
3. **Compare to standard error:** $\text{SE}(\Delta\text{WAIC})$

Decision rule:

If $|\Delta\text{WAIC}| < 2 \times \text{SE}$: Models equivalent
If $|\Delta\text{WAIC}| > 2 \times \text{SE}$: Models differ significantly

Example:

- Model 1 (linear): $\text{WAIC} = 245.3 \pm 12.1$
- Model 2 (quadratic): $\text{WAIC} = 238.7 \pm 11.8$
- Difference: $\Delta\text{WAIC} = 6.6$, $\text{SE}(\Delta) \approx 2.3$
- $6.6 > 2 \times 2.3 \rightarrow$ Model 2 preferred

WAIC: Numerical Example

Data (5 weeks)

Week	Sales	Spend	Season
1	100	10	0.0
2	120	15	0.5
3	110	12	1.0
4	105	11	0.5
5	115	14	0.0

Model 1: $y = \beta_0 + \beta_1 x$ (3 params: β_0, β_1, σ)
Model 2: $y = \beta_0 + \beta_1 x + \gamma \cdot s$ (4 params: $+\gamma$)

Posterior Samples (Model 1, $S = 3$)

Sample	β_0	β_1	σ
1	90	1.20	5
2	92	1.10	5
3	91	1.15	5

Step 1a: Predictions for Week 1 ($y_1 = 100, x_1 = 10$)

Sample	$\hat{y}_1 = \beta_0 + \beta_1 \times 10$	$p(y_1 \mid \theta^{(s)})$
1	$90 + 1.20 \times 10 = 102$	0.0782
2	$92 + 1.10 \times 10 = 103$	0.0727
3	$91 + 1.15 \times 10 = 102.5$	0.0761
Mean likelihood		0.0757
$\text{lppd}_1 = \log(0.0757)$		**-2.58**

Note: $p(y_1 \mid \theta^{(s)}) = \mathcal{N}(y_1; \hat{y}_1, \sigma = 5)$

Step 1b: Complete lppd Calculation

Week	Mean $p(y_i \mid \theta)$	$\log(\cdot)$	lppd_i
1	0.0757	log(0.0757)	-2.58
2	0.0442	log(0.0442)	-3.12
3	0.0663	log(0.0663)	-2.71
4	0.0799	log(0.0799)	-2.52
5	0.0556	log(0.0556)	-2.89
Model 1 Total lppd			**-13.82**

Step 2a: Variance Calculation for Week 1

Sample	$\log p(y_1 \mid \theta^{(s)})$
1	$\log(0.0782) = -2.548$
2	$\log(0.0727) = -2.621$
3	$\log(0.0761) = -2.575$
Mean	-2.581
Variance	$((-2.548 + 2.581)^2 + \cdots)/2$
$p_{\text{WAIC},1}$	**0.00062**

Step 2b: Complete p_{WAIC} (Model 1)

Week	$\text{Var}[\log p(y_i \mid \theta)]$
1	0.00062
2	0.00143
...	...
$\Sigma\ p_{\text{WAIC}}$	**0.00443**

WAIC: Numerical Example

Step 3: Pointwise WAIC Calculation

Week	lppd_i	$p_{\text{WAIC},i}$	WAIC_i
Model 1:			
1	-2.58	0.00062	5.162
2	-3.12	0.00143	6.243
3	-2.71	0.00089	5.422
4	-2.52	0.00051	5.041
5	-2.89	0.00098	5.782
Total WAIC_1			**27.650**
Model 2:			
1	-2.41	0.00189	4.826
2	-2.89	0.00412	5.788
3	-2.53	0.00298	5.066
4	-2.38	0.00167	4.763
5	-2.24	0.00321	4.487
Total WAIC_2			**24.940**

Formula: $\text{WAIC}_i = -2(\text{lppd}_i - p_{\text{WAIC},i})$

Step 4: Calculate SE(ΔWAIC)

Week	$\text{WAIC}_{1,i}$	$\text{WAIC}_{2,i}$	Δ_i
1	5.162	4.826	0.336
2	6.243	5.788	0.455
3	5.422	5.066	0.356
4	5.041	4.763	0.278
5	5.782	4.487	1.295
$\Delta\text{WAIC} = \sum \Delta_i$			**2.710**
$\text{Mean}(\Delta_i)$			0.542
$\text{Var}(\Delta_i)$			0.180
$\text{SE}(\Delta) = \sqrt{n \times \text{Var}}$			**0.950**

Step 5: Statistical Decision

Metric	Value
$\lvert\Delta\text{WAIC}\rvert$	2.710
SE(ΔWAIC)	0.950
Threshold (2× SE)	1.900
Is $\lvert\Delta\rvert >$ Threshold?	**YES**

LOO-CV: Numerical Example

Data (5 weeks)

Week	Sales	Spend	Season
1	100	10	0.0
2	120	15	0.5
3	110	12	1.0
4	105	11	0.5
5	115	14	0.0

Concept: Fit model on $\mathbf{y}_{-i}$, predict left-out y_i
Model 1: $y = \beta_0 + \beta_1 x$
Model 2: $y = \beta_0 + \beta_1 x + \gamma \cdot s$

Step 1a: Leave Out Week 1 (Model 1)

Training: Weeks {2,3,4,5}

Week	Sales	Spend
2	120	15
3	110	12
4	105	11
5	115	14

Fitted: $\hat{\beta}_0 = 92$, $\hat{\beta}_1 = 1.05$

Prediction for Week 1:

Predict: $\hat{y}_1 = 92 + 1.05 \times 10 = 102.5$
Actual: $y_1 = 100$
Error: $100 - 102.5 = -2.5$

$p(y_1|\mathbf{y}_{-1}) = \mathcal{N}(100; 102.5, 5) = 0.0761$
$\log p(y_1|\mathbf{y}_{-1}) = \mathbf{-2.61}$

Step 1b: Complete LOO Process (Model 1)

Left Out	Trained on	Prediction Pred	Prediction Actual	log p $(y_i\|\mathbf{y}_{-i})$
1	{2,3,4,5}	102.5	100	-2.61
2	{1,3,4,5}	108.8	120	-3.21
3	{1,2,4,5}	105.2	110	-2.79
4	{1,2,3,5}	104.0	105	-2.55
5	{1,2,3,4}	107.1	115	-2.92
ELPD_{LOO} (M1)				**-14.08**

Step 1c: Complete LOO Process (Model 2)

Left Out	Prediction Pred	Prediction Actual	log p $(y_i\|\mathbf{y}_{-i})$
1	100.8	100	-2.44
2	119.2	120	-2.93
3	109.5	110	-2.59
4	104.6	105	-2.41
5	114.3	115	-2.31
ELPD_{LOO} (M2)			**-12.68**

Note: Model 2 makes better predictions (closer to actual, higher log p)
Problem: Refitted model 5 times per model!
Solution: ???

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Summary

Key concepts covered today:

- **Bayesian regression**: Prior $\times$ Likelihood $\rightarrow$ Posterior distribution over β
- **Priors as regularization**: Gaussian prior = Ridge
- **Full uncertainty**: Not just point estimates, but distributions
- **Model checking**: PPC, residuals, WAIC/LOO

Workflow:

1. Specify model (priors + likelihood) in PyMC
2. Sample posterior (MCMC with diagnostics)
3. Check model fit (PPC, residuals)
4. Compare models (WAIC/LOO)
5. Sensitivity analysis
6. Make predictions with uncertainty

Next Session: Sensitivity & Robustness Analysis

Global and Local Sensitivity:

- How do the results change when changing the parameter values?
- How do the results change when changing **a set of** parameters?

Robustness:

- What if we change the model's assumptions/priors?
- What if we change the model's structure?

Questions?

Next: Sensitivity Analysis!